Who bears the tariff? The 2025 US–UK trade shock through the tax–benefit system*

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Abstract

I study who bears the 2025 US tariffs on UK goods after the tax–benefit system responds. The shock enters through primitives: the tariff schedule (10 per cent baseline, 25 per cent autos and steel, later mitigated by the Economic Prosperity Deal) is combined with US-export intensity by SIC division—built from HMRC trade-by-SIC and ONS turnover data—and an export-demand elasticity of 2.0, anchored on the realised outturn (US-bound goods exports fell 24.7 per cent in April 2025 against a 12.8 per cent trade-weighted tariff), to derive sector-level earnings shocks passed through PolicyEngine UK on Family Resources Survey 2024–25 microdata. Employment is an outcome of the derived shock, not a free input. Holding the aggregate gross earnings loss fixed (£1.8 billion a year, 0.06 per cent of GDP on the direct channel), I vary the *adjustment margin*: job loss, wage cuts with sectoral reallocation, or exit into economic inactivity. The headline finding reverses the naive prior: the tax–benefit system cushions in-work wage cuts (41 per cent) *more* than earnings-equivalent job losses (33 per cent), and an exact component decomposition demonstrates why: income tax relieves wage cuts at marginal rates but job losses only at average rates (35 versus 20 pence per pound of gross loss), while Universal Credit reaches only a third of displaced workers' families, gated by modelled take-up and the joint partner-income taper. The shock reaches households mainly as a fiscal event—£0.7–1.0 billion a year to the Exchequer (Monte Carlo means over 50 displacement draws, draw-to-draw standard deviation £0.3 billion) against a poverty rise of at most 0.06 percentage points—and, on the displacement margin, is regionally rather than vertically concentrated: losses land in deciles 7–10, and while national inequality statistics barely move, the hardest-hit regions average £60–110 per person per year across draws, with any single realisation concentrating £100–200 per head on a few regions. The Economic Prosperity Deal averted roughly 6,900 job losses in expectation, £0.4 billion of gross earnings and roughly £0.2 billion a year of Exchequer cost (positive in 36 of 50 paired draws).

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1 Introduction

Who bears a foreign tariff shock after the tax–benefit system responds—and how much does the answer depend on whether workers adjust through job loss, wage cuts with sectoral reallocation, or exit into economic inactivity? In April 2025 the United States imposed a 10 per cent baseline tariff on UK goods, with 25 per cent rates on cars and steel; the May 2025 Economic Prosperity Deal (EPD) cut the auto rate to 10 per cent on a 100,000-vehicle quota, offered conditional steel relief, and—in a December 2025 side agreement paid for through NHS pricing concessions—exempted pharmaceuticals. A rich institutional literature has traced the macroeconomic, sectoral, regional and firm-level consequences of this shock (OBR, 2025; Bank of England, 2025; City-REDI, 2025; Centre for Cities, 2025). None of it reaches the household: no analysis passes the tariff through to disposable income, poverty, or the benefit system that would actually cushion it. This paper fills that cell.

The design is primitives-first. The shock enters as the tariff schedule itself, combined with US-export intensity by SIC division (ONS/HMRC trade-by-SIC) and an export-demand elasticity, to yield sector-level earnings shocks; the elasticity (2.0) is anchored on the realised outturn—US-bound goods exports fell 24.7 per cent in the first full tariff month against a 12.8 per cent trade-weighted tariff—and cross-checked against UK worker-level trade-shock estimates (De Lyon and Pessoa, 2021) and the OBR and Ignatenko et al. (2025) aggregate anchors. Employment is an *outcome* of the derived shock, not a free input. The derived sector shocks are then realised on Family Resources Survey workers in exposed industries under three adjustment-margin families, each holding the aggregate earnings loss fixed: *displacement* into unemployment (the China-shock short run of Autor et al., 2013), *wage cuts with reallocation* into other sectors (the German and Danish long run of Dauth et al., 2021 and Traiberman, 2019), and *exit into inactivity* for older displaced workers—the margin UK deindustrialisation actually chose, when coalfield and steel job losses became benefit-financed economic inactivity rather than measured unemployment (Beatty et al., 2007; Beatty and Fothergill, 2017). PolicyEngine UK, an open-source tax–benefit microsimulation model, computes disposable income, poverty, inequality and the Exchequer position under each.

The adjustment margin is the paper's central axis, and the results reverse the prior that motivated it. The naive reading of Universal Credit says the system should replace a job loss more generously than it tops up an in-work wage cut of equal aggregate size. On the microdata the opposite holds: the tax–benefit system cushions 41 per cent of the shock when it arrives as wage cuts but only 33 per cent when it arrives as displacement. An exact component decomposition demonstrates why: a wage cut is relieved at every worker's *marginal* income-tax rate (35 pence per pound of gross loss) while a job loss destroys tax liability only at the *average* rate (20 pence)—a gap that alone exceeds the headline difference—and Universal Credit, though it responds more to displacement than to wage cuts, reaches only a third of displaced workers' families, gated in the model by take-up and the joint partner-income taper (the capital means test cannot bind on FRS data, which carry no household capital). In the language of Atkin et al. (2025), statutory generosity toward the unemployed is not economic generosity once the gates bind on the population actually displaced. Whichever margin, the calibrated shock (a £1.8 billion annual gross earnings loss, 0.06 per cent of GDP on the direct channel) reaches households mainly as a fiscal event—£0.7–1.0 billion a year to the Exchequer (Monte Carlo means over 50 displacement draws)—rather than a poverty event (at most +0.06 percentage points); and on the displacement margin it is regionally, not vertically, concentrated, with losses landing in deciles 7–10 of the income distribution while averaging £60–110 per person per year in the hardest-hit regions—and reaching £100–200 per head in the few regions any single realisation lands on. Which margin the data chose is testable against the 2025–26 outturn (ONS trade, BICS, SMMT monthly series), and the historical UK evidence decomposing displacement costs into wage cuts versus non-employment spells (Rud et al., 2022) calibrates the split.

Two sectoral nuances discipline the headline claims. Steel carries a caveat: roughly 80 per cent of UK steel exports go to the EU, so the steel–Wales cell is more exposed to EU/UK safeguards than to US tariffs. And pharmaceuticals—exempted under the December 2025 deal in exchange for VPAG rebate cuts and NHS pricing concessions—is a fiscal and pricing channel, not an employment channel, and is treated as such. Finally, an EPD counterfactual (full tariffs versus deal-mitigated rates) prices what the deal was worth, sector by sector and region by region, to households and the Treasury. Following Ignatenko et al. (2025), revenue recycling can flip aggregate employment signs; we compute first-round fiscal incidence holding fiscal policy fixed.

The remainder of the paper proceeds as follows. Section 2 situates the study between the quantitative trade-incidence literature and tax–benefit microsimulation. Section 3 sets out the tariff schedule, the trade-by-SIC exposure mapping, the derived sector shocks and the three adjustment-margin families. Section 4 presents the results: the derived shocks and their outturn anchor, the cushioning reversal, the distributional, fiscal and regional incidence by margin, and the EPD counterfactual. Section 5 turns to the EPD counterfactual and policy responses. Section 6 concludes.

2 Literature

This paper sits at the intersection of four literatures: the quantitative and reduced-form work on the incidence of trade shocks; the evidence on how workers actually adjust to them; the institutional analyses of the 2025 US tariffs on the UK; and the microsimulation literature on tax–benefit cushioning of earnings shocks. The intersection itself is empty: no study passes a trade-policy shock through a full tax–benefit microsimulation to household disposable income, poverty and the Exchequer.

2.1 Trade shocks and their incidence

Two recent papers anchor the structural end and frame the adjustment-margin question as its poles. Ignatenko et al. (2025) build a quantitative general-equilibrium model—nesting Armington, Eaton–Kortum, Krugman and Melitz structures with exact hat algebra over 194 countries—in which the 2025 tariffs enter as a *primitive* on trade shares. Wages are flexible and labour supply elastic, so employment is an equilibrium *outcome*; strikingly, its sign flips with revenue recycling (a tariff-financed income-tax cut raises US employment by 0.32 per cent, a lump-sum rebate lowers it by 0.41 per cent), and the optimal US tariff is a uniform 19 per cent independent of bilateral deficits. The UK is in their sample, and their replication data provide the UK-specific welfare and employment cells we use as aggregate anchors. Atkin et al. (2025), by contrast, apply a Baqaee–Farhi wedge framework to Chilean administrative data with factor supplies *fixed*: there is no employment margin at all, all adjustment runs through flexible wages and prices, and household incidence arrives via consumption baskets, factor ownership, profits and transfers—their “statutory versus economic incidence” framing is the one we adopt. One paper derives employment from the tariff primitive; the other assumes it fixed and lets wages absorb everything. A third structural anchor sits between the poles: Rodríguez-Clare et al.

(2025) embed the 2025 trade war in a dynamic trade-and-reallocation model with *downward nominal wage rigidity*, in which higher tariffs raise US manufacturing employment but lower total employment as falling real wages depress participation, with US real income down about 1 per cent by 2028—rigid wages turn the same primitive into an employment shock that flexible-wage models turn into a wage shock, which is precisely the disagreement our adjustment-margin axis makes an explicit scenario dimension. On the partner-country side, Michelena et al. (2025) run the 2025 escalation through a multiregional input–output framework and find employment and export losses concentrated in the *targeted* countries—the same statutory-versus-economic distinction at the country level, but at the level of sectoral aggregates rather than households. Complementing the model-based work, den Besten and Känzig (2025) construct a narrative series of US tariff shocks over 1840–2024 and find tariff increases robustly contractionary—imports fall sharply, exports decline with a lag, and output drops persistently—historical discipline for treating the shock as a negative export-demand shock rather than a benign relative-price change.

The reduced-form lineage begins with Autor et al. (2013): import competition caused large, persistent employment and wage losses in exposed US commuting zones, with effects persisting to 2019 and no offsetting migration (Autor et al., 2021). Caliendo et al. (2019) provide the structural counterpart—a dynamic spatial model with costly mobility in which the China shock is a productivity/trade-cost primitive, generating aggregate gains alongside concentrated regional losses—and Adão et al. (2023) supply the theory-consistent mapping from regional shift-share estimates to aggregate counterfactuals that cautions against reading regional results as national ones. Kim and Vogel (2021) translate reduced-form employment and wage responses into money-metric worker welfare, finding regressive incidence with displaced workers' losses exceeding wage losses; Galle et al. (2023) show in a Roy-model setting that aggregate gains coexist with group-specific losses. Our contribution relative to this structural incidence work is to replace stylised worker groups and stylised transfers with real households and the actual UK tax–benefit system.

The 2018–19 US tariff episode supplies the sharpest direct evidence on who pays a tariff. Pass-through into US import prices was essentially complete, placing the incidence on the importing country's firms and consumers (Amiti et al., 2019; Fajgelbaum et al., 2020), with retail prices adjusting more slowly than border prices (Cavallo et al., 2021)—a pattern the 2025 wave repeats: matching daily retailer microdata to product-level tariff rates, Cavallo et al. (2025) estimate retail pass-through of about 20 per cent by September 2025, contributing roughly 0.7 percentage points to the US CPI; and Flaaen and Pierce (2019) show that US manufacturing employment fell on net, as input-cost increases and foreign retaliation outweighed import protection. The retaliation side of that episode is the closest analogue to our channel—a partner's tariff arriving as an export-demand shock: Waugh (2019) finds Chinese retaliatory tariffs depressed consumption in exposed US counties, and Blanchard et al. (2019) trace the electoral fallout of retaliatory tariffs across US counties. The broader distributional literature—surveyed in Fajgelbaum and Khandelwal (2016) and Fajgelbaum and Khandelwal (2022), with Borusyak and Jaravel (2021) decomposing expenditure against earnings channels—establishes that trade shocks redistribute along both consumption and income margins. All of this work prices tariffs at the border, the store, or the county; none of it passes the shock through a household-level tax–benefit system, which is the step taken here. Two propagation channels our direct-channel design deliberately excludes have well-measured analogues in this literature. Firm-level shocks travel along supply chains—Barrot and Sauvagnat (2016) show idiosyncratic disruptions imposing substantial output losses on customers of affected suppliers, and Acemoglu et al. (2016) find network effects on aggregate outcomes several times larger than the direct effect—and tradable-sector job losses multiply into local services, with each lost tradable job costing roughly 0.4–1.6 additional local jobs across the US and German estimates (Moretti, 2010; Gathmann et al., 2020). Both channels scale our direct-channel numbers upward; Section 6 carries the caveat. Open-economy macro adds an offsetting margin: exchange rates absorb part of a tariff, but imperfectly and state-dependently—at most a fifth of the 2018–19 dollar appreciation is attributable to the tariffs (Jeanne and Son, 2024), the dollar appreciates on unilateral tariffs but depreciates under retaliation (Corsetti et al., 2025), and tariff *uncertainty* itself depreciated the dollar in 2025 through a risk-premium channel (Kalemli-Özcan et al., 2026); model-based analyses of retaliation similarly find non-trivial consumption losses for passive trade partners (Auray et al., 2020). We hold the exchange rate fixed and let the outturn-anchored elasticity absorb whatever FX offset in fact occurred. Early assessments of the 2025 wave itself (Fajgelbaum and Khandelwal, 2026) again find the incidence landing on the tariff-imposing country's consumers—complementary to our question, which is the incidence on the tariffed *partner's* workers.

2.2 Adjustment margins: how workers absorb trade shocks

The worker-level evidence says the margin is mostly earnings, not permanent unemployment. Autor et al. (2014) find US workers exposed to import competition lose earnings amid churn across employers and sectors rather than through indefinite joblessness. Traiberman (2019) shows for Denmark that occupation-specific human capital produces reallocation with persistent wage losses; because our shock is an *export-demand* shock rather than import competition, the most direct microeconometric precedents are Hummels et al. (2014), who instrument Danish firm-level export demand and trace it into worker wages by task, and Garin and Silvério (2024), who show that Portuguese exporters pass idiosyncratic export-demand shocks into the wages of incumbent workers rather than shedding them—firm-level evidence for exactly the wage-cut family of our scenario grid. Dauth et al. (2021) provide the cleanest evidence for the wage-margin family, with German workers reallocating into services and other plants at lower wages. Dix-Carneiro (2014) estimates that reallocation takes years, with adjustment costs of 1.4–2.7 times annual wages and older workers hit hardest—calibrating our transition horizons and age gradients—while Dix-Carneiro and Kovak (2017) show regional effects that *grow* over two decades on a joint wage/employment margin.

The UK's own history points to a third margin absent from the US and German studies: benefit-financed economic inactivity. Beatty et al. (2007) document that coalfield job losses became male inactivity on incapacity benefits, not measured unemployment, with incomplete recovery after twenty years; Beatty and Fothergill (2017) show that 1980s–90s industrial job destruction *still* inflates the deficit through higher benefit claims and lower tax receipts—the direct UK precedent for our fiscal-incidence question, retrospective and area-level where ours is prospective, rules-based and household-level. The margin is not merely historical: Roberts and Taylor (2022) show in 2009–18 longitudinal data that disability benefit claims still respond to local labour-market slack over and above health status, Beatty and Fothergill (2023) document the persistence of hidden unemployment among incapacity claimants across large parts of Britain, and the IFS records 4.2 million working-age people on health-related benefits by 2024, up from 3.2 million in 2019 (IFS, 2024)—the inactivity margin is live at exactly the moment the tariff shock arrives. Aragón et al. (2018) add gender-differentiated margins from mine closures, and Rud et al. (2022) decompose displacement costs into wage cuts on re-employment versus non-employment spells—directly calibrating the split between our displacement and wage-cut families. For UK-specific elasticities, De Lyon and Pessoa (2021) is the key source: UK workers in Chinese-import-competing industries lost employment years and earnings without equally gainful re-employment, with larger female losses through employment years (see also Dorn and Levell, 2024). Closest to our object, Irastorza-Fadrique et al. (2025) study UK *household* responses to trade shocks—added-worker effects, later retirement, self-employment—but model behaviour, not tax–benefit mechanics; we treat their margins as complementary to our rules-based incidence.

2.3 The 2025 tariffs and the UK

The institutional literature on the 2025 shock is rich but stops above the household. NIESR's NiGEM scenarios and the OBR's March 2025 tariff analysis provide the macro anchors—the OBR puts the permanent UK GDP loss from a 20-point US tariff at one-third to three-quarters of a per cent (NIESR, 2024; OBR, 2025). The Bank of England estimates the effective US tariff on the UK at roughly 9 per cent (against about 18 per cent globally) and finds trade diversion has mildly *lowered* UK import prices (Bank of England, 2025; Dhingra, 2025)—which is what licenses our isolating the export-demand/earnings channel while holding the price and monetary channels fixed; the Bank's own staff work on the exchange-rate response to tariffs and retaliations (Corsetti et al., 2025) supplies the FX side of that held-fixed channel. The multilateral institutions frame the aggregates: the IMF's October 2025 outlook has global growth slowing to 3.2 per cent in 2025 amid the tariff shock (IMF, 2025), and the WTO's October 2025 update puts merchandise trade volume growth at 2.4 per cent in 2025 but only 0.5 per cent in 2026 as the tariffs bind (WTO, 2025). Firm surveys find most UK firms expecting small negative effects, with the US taking about 22 per cent of UK exports (CEP, 2025). The regional and sectoral geography is sharp: Coventry sends 22 per cent of its exports to the US (Centre for Cities, 2025); auto tariffs are estimated to cost £10bn of GDP over five years with the West Midlands bearing 62 per cent of it (City-REDI, 2025); SMMT data show US vehicle exports collapsing from 7,077 units in April to 4,223 in May 2025 before the EPD's implementation. UK Steel notes that some 80 per cent of UK steel exports go to Europe—the basis of our steel caveat—and the December 2025 pharmaceutical deal (0 per cent tariffs for three years, paid for through VPAG rebate cuts from 22.5 to 14.5 per cent and NHS pricing concessions) makes pharma a fiscal/pricing shock rather than an employment shock (ABPI, 2025). The Resolution Foundation's household-facing analysis is descriptive rather than microsimulated (Resolution Foundation, 2025a); the IFS has no dedicated distributional publication on the 2025 tariffs. Across macro, sector, region and firm, the household/fiscal cell is empty.

For UK trade-shock precedents more broadly, Dhingra et al. (2017) map Brexit trade costs to income losses, and Fetzer and Wang (2024) provide the best district-level trade-shock incidence estimates (synthetic-control output losses of 5–10 points concentrated in manufacturing districts); Fetzer (2019) and Bloom et al. (2019) supply the political-economy and uncertainty channels.

2.4 Tax–benefit cushioning of earnings shocks

The methodological ancestors are Dolls et al. (2012), who use EUROMOD and TAXSIM to show EU tax–benefit systems absorb about 38 per cent of an income shock and 47 per cent of an unemployment shock (against roughly 32 per cent in the US), and Sutherland and Figari (2013) on EUROMOD's stress-test use—but both apply stylised macro shocks, not trade-policy-derived ones. Our direct methodological sibling is Brewer and Tasseva (2021), who pass the Covid-19 labour-market shock through UKMOD to show that the UK tax–benefit system plus the emergency support schemes held the rise in income inequality far below the rise in earnings inequality: the same architecture—an externally derived shock realised on survey households and run through the full UK benefit rules—applied to a pandemic where ours applies it to a trade-policy shock, without the emergency schemes and with the adjustment margin as the central axis. The wider macro-micro linkage literature supplies the lineage for the shock-derivation step itself: Bourguignon et al. (2008) codify the techniques for passing macro shocks to household microdata, Peichl (2016) surveys the linkage of microsimulation to structural macro models, and Navicke et al. (2013) show how EUROMOD nowcasts impose estimated labour-market transitions on survey microdata—the same device as our displacement draws, applied there to observed rather than counterfactual shocks. On the US side, exposed commuting zones saw rising disability, retirement and welfare payments that offset only a small fraction of earnings losses (Autor et al., 2013); Hyman et al. (2024) show wage insurance raises re-employment and earnings, a natural counterfactual reform for our framework. The nearest tariff-microsimulation links are on the US side: the Tax Policy Center's tariff models allocate tariff burdens to tax units through consumption baskets in a full microsimulation model (Tax Policy Center, 2025), the Budget Lab at Yale prices the 2025 schedule's fiscal and distributional effects by income decile (Budget Lab, 2025), and Acosta and Cox (2024) document that the US tariff code is built regressively, with higher rates on lower-end varieties of the same goods. Two institutional facts underpin the cushioning asymmetry our results turn on. UC's capital rules—thresholds frozen in cash terms since 2006—reduced or extinguished entitlement for around two million income-eligible families, with roughly 1.1 million standing to gain from uprating alone (Resolution Foundation, 2025b); and take-up of income-related benefits is well below 100 per cent (DWP, 2024), so statutory replacement rates overstate what displaced workers actually receive. Both push in the direction of our finding that the means-tested extensive margin cushions less than the automatic in-work margin—take-up gates entitlement inside the model, while the capital rules are a real-world gate the plain FRS data cannot exercise (Section 6). But all of these allocate the *consumer-price* incidence of a country's own tariffs; ours is the export-demand/earnings incidence of a *partner's* tariff. The distinction between these channels, and the absence of any analysis in ours, defines the gap this paper fills: structural incidence models have no realistic fiscal system, microsimulation has no trade shock, and the adjustment-margin axis—which has first-order fiscal and distributional consequences because Universal Credit replaces unemployment far more generously than in-work wage cuts—has not been quantified by anyone.

3 Methodology

3.1 Design: primitives first

The shock enters through primitives, not through an assumed employment effect. The pipeline is:

$$s_j \;=\; \tau_j \, x_j \, \varepsilon \, \pi, \tag{1}$$

where s_j is the derived earnings-shock size for SIC 2007 division j (the fraction of the division's employee wage bill removed by the tariff), τ_j is the tariff rate faced by the division under the schedule, x_j is its US-export intensity, ε is the export-demand elasticity, and π is the pass-through of the sector output loss to the sector wage bill. x_j is US goods exports of the division as a share of division turnover, built from the HMRC uktradeinfo RTS/OTS API (2024 US exports, £52.5 billion mapped to SIC divisions of a £55.6 billion customs-basis total, via a documented SITC-to-SIC crosswalk) and ONS Annual Business Survey turnover denominators; Appendix A gives the full provenance and validation. The elasticity $\varepsilon = 2.0$ is anchored on the realised outturn: ONS data show UK goods exports to the US falling 24.7 per cent in April 2025, the first full tariff month, against a 12.8 per cent trade-weighted average tariff implied by the schedule and the built intensities ($0.247/0.128 = 1.93$, rounded to 2.0, mid-range of short-run trade elasticities). The pass-through $\pi = 1.0$ places the full sector output loss on the sector wage bill—the appropriate short-run displacement assumption. Because ε and π enter Equation (1) only through their product, all pound-denominated results scale approximately linearly in $\varepsilon\pi$, while cushioning *rates* and margin orderings are essentially invariant to it; Appendix D verifies both properties on a 4×3 sensitivity grid over $\varepsilon \in \{1.0, 1.5, 2.0, 3.0\}$ and $\pi \in \{0.5, 0.75, 1.0\}$. Since the central $\varepsilon = 2.0$ is anchored on a single month's realised outturn (April 2025), the grid doubles as the uncertainty band around that anchor. The implied aggregate gross earnings loss under the full schedule is £1.77 billion a year, about 0.06 per cent of GDP on the direct exposed-sector channel, sensibly below the OBR's all-channel scenario range of 0.3–0.6 per cent (OBR, 2025) and consistent with the UK cell of Ignatenko et al. (2025) and UK worker-level responses (De Lyon and Pessoa, 2021; Rud et al., 2022). Employment is then an *outcome* of s_j under an assumed adjustment margin—never a free input. The design covers the direct exposed-sector channel only: local-multiplier estimates of 0.4–1.6 additional local jobs per lost tradable job (Moretti, 2010; Gathmann et al., 2020) imply the all-channel employment effect is larger, so the derived shock is a lower bound (Section 6).

3.2 The tariff schedule and the EPD counterfactual

Two tariff scenarios bracket the policy. *Full tariffs*: the April 2025 schedule—10 per cent baseline on UK goods, 25 per cent on motor vehicles (SIC 29) and steel (SIC 24). *EPD*: the deal-mitigated schedule—autos at the 10 per cent in-quota rate (100,000 vehicles, roughly the 2024 US export volume, so the quota binds only marginally); steel with conditional, partial relief; pharmaceuticals (SIC 21) exempt under the December 2025 agreement. The difference between the two scenarios, run through identical downstream machinery, prices the EPD for households and the Exchequer. Two caveats travel with the schedule. Steel: about 80 per cent of UK steel exports go to the EU, so the US tariff is a minority driver of the sector's exposure and the steel cell should be read alongside EU/UK safeguards. Pharmaceuticals: the exemption was purchased through VPAG rebate cuts and NHS pricing concessions, so pharma's incidence is fiscal and price-side, not an employment shock; we model its employment shock as zero under the EPD and discuss the fiscal channel separately (Section 5).

3.3 Adjustment-margin families

The derived s_j is realised on Family Resources Survey 2024–25 employees in exposed divisions under three families, each holding the aggregate earnings loss $\sum_j s_j B_j$ (with B_j the division wage bill) fixed:

Displacement (ADH short run). Within each exposed division a weighted head-count quota $s_j\times$ (division employee weight) is drawn with uniform ordering keys, so a represented worker's displacement probability does not depend on their record's grossing weight; displaced workers move to unemployment with earnings, hours, pension contributions and statutory pay zeroed.

Wage cuts with reallocation (Dauth/Traiberman long run). No job loss; every employee in division j takes a proportional cut of s_j, so the weighted aggregate earnings removed equals $\sum_j s_j B_j$ by construction—exact earnings-equivalence with the displacement family's expected removal.

Inactivity (Beatty–Fothergill UK history). The displacement draw, but displaced workers aged 50 and over exit to economic inactivity, the benefit-financed route of UK deindustrialisation, rather than unemployment. In the current model version (2.89.2 of policyengine-uk) the status transition is applied and verified, but the UC health element and conditionality are not modelled, so computed entitlements—and hence all downstream results—coincide with the displacement family's; the margin is retained as scoped machinery for the health-element extension (Section 4.4).

The margin is the paper's central axis because the tax–benefit system treats the two poles asymmetrically: an in-work earnings reduction is offset automatically at marginal tax and taper rates, while a job loss moves the worker to the gated, means-tested extensive margin of the benefit system. Which pole the system cushions more is an empirical question the microdata answer—in the direction opposite to the naive replacement-rate intuition (Section 4.2).

3.4 Microsimulation and outcomes

The realisation step—an externally derived shock imposed as status and earnings transitions on survey households, then run through the full benefit rules—follows Brewer and Tasseva (2021), who apply it to the Covid-19 labour-market shock in UKMOD, and the EUROMOD nowcasting tradition of imposed labour-market transitions (Navicke et al., 2013). PolicyEngine UK computes baseline and shocked household disposable income (HBAI cash concept throughout), relative before-housing-costs poverty, absolute before-housing-costs poverty against a fixed baseline line (computed manually, since the installed model version exposes no absolute-poverty variable; Appendix C), after-housing-costs poverty, the Gini coefficient of equivalised disposable income, decile and region breakdowns, and the Exchequer effect (foregone tax plus additional benefit spending, via the government balance). Displacement draws are Monte Carlo: every displacement/inactivity result is reported as the mean and standard deviation over n draws (default 20). Following Ignatenko et al. (2025), revenue recycling can flip aggregate employment responses; we compute first-round incidence holding fiscal policy fixed and treat recycling as a policy counterfactual.

4 Results

All results are computed on Family Resources Survey 2024–25 microdata for the 2026 period, on the HBAI cash disposable income concept throughout (Section 3.4). Displacement and inactivity results are means over 50 Monte Carlo draws with $\pm$ values denoting standard deviations across draws; the wage-cut margin is deterministic. Decile and regional profiles are likewise Monte Carlo means across the 50 draws. Because the draw-to-draw standard deviation on the Exchequer cost is large (about £0.30 billion against means of £0.7–0.9 billion), I narrate only orderings and contrasts that survive it, and flag those that do not.

I lead with the paper's headline finding, because it reverses the prior that motivated the design. Section 3.3 argued—as the naive reading of Universal Credit suggests—that the tax–benefit system should replace a job loss more generously than an in-work wage cut of equal aggregate size. On the FRS microdata the opposite holds: the system cushions 33 per cent of the gross earnings loss when the shock arrives as displacement, but 41 per cent when the identical aggregate loss arrives as proportional wage cuts. Section 4.3 decomposes the gap component by component. Two mechanisms carry it. First, tax arithmetic: a wage cut is relieved at every affected worker's *marginal* income-tax rate (35 pence per pound of gross loss), whereas a job loss wipes out the whole earnings stream—personal allowance and all—so the income-tax offset falls to the *average* rate (20 pence). Second, the extensive margin of the benefit system gates out-of-work support: two-thirds of displaced workers' families receive no UC at all post-shock, so UC restores only 4 pence per pound of displacement loss—though the binding gates in the model are take-up and the joint partner-income taper, not the capital test, which the FRS data cannot exercise. The adjustment margin therefore determines not just *where* the shock lands but *how much of it* the state absorbs—and in the direction opposite to the standard intuition.

4.1 The derived sector shocks

Table 1 reports the primitives and the derived shock for the most exposed SIC 2007 divisions. US-export intensity—2024 US goods exports over division turnover, built from the HMRC RTS/OTS API and the ONS Annual Business Survey as described in Appendix A—is highest in machinery (0.180), computer/electronic and optical products (0.179), other transport equipment including aerospace (0.157), pharmaceuticals (0.130), electrical equipment (0.126), chemicals (0.125) and motor vehicles (0.110); basic metals including steel, despite its 25 per cent tariff, has intensity of only 0.051 because most UK steel output does not go to the US. With the calibrated elasticity $\varepsilon = 2.0$ and full wage-bill pass-through (Section 3.1), the derived earnings shocks $s_j = \varepsilon\tau_j x_j$ under the full schedule are led by autos (5.5 per cent of the division wage bill), machinery (3.6 per cent), electronics (3.6 per cent), aerospace/other transport (3.1 per cent), pharmaceuticals (2.6 per cent) and steel (2.5 per cent). Under the EPD, the auto shock falls to 2.2 per cent, steel to 1.3 per cent, and pharmaceuticals to zero.

The anchor check. Applied to FRS 2024–25 employees, the full schedule removes £1.77 billion of gross annual earnings—about 0.06 per cent of GDP. This is the *direct exposed-sector earnings channel only*, and it sits sensibly below the OBR's March 2025 all-channel scenario range of roughly 0.3–0.6 per cent of GDP (OBR, 2025), which additionally includes uncertainty, investment and general-equilibrium channels that a static microsimulation deliberately excludes. Under the EPD schedule the gross earnings loss is £1.39 billion. The elasticity itself is anchored on the realised outturn rather than borrowed from the literature: UK goods exports to the US fell 24.7 per cent in April 2025 against a trade-weighted average full-schedule tariff of 12.8 per cent computed from Table 1's primitives, giving $0.247/0.128 = 1.93$, rounded to 2.0—also mid-range of short-run trade elasticities (Section 4.9). Because the elasticity and the pass-through enter the derived shock only through their product, the pound levels above scale approximately linearly in $\varepsilon\pi$ while every rate and ordering in this section is essentially invariant to it: Appendix D verifies this on a 4×3 grid over $\varepsilon \in \{1.0, 1.5, 2.0, 3.0\}$ and $\pi \in \{0.5, 0.75, 1.0\}$, which also serves as the uncertainty band around the one-month outturn anchor.

Realised as displacement, the full schedule puts 34,200 workers (±3,600 across 50 draws) out

Table 1: Primitives and derived earnings shocks, most exposed SIC divisions

SIC	Division	Tariff τ_j Full	Tariff τ_j EPD	US-export intensity x_j	Derived shock s_j (%) Full	Derived shock s_j (%) EPD
29	Motor vehicles	0.25	0.10	0.110	5.5	2.2
28	Machinery and equipment	0.10	0.10	0.180	3.6	3.6
26	Computer, electronic, optical	0.10	0.10	0.179	3.6	3.6
30	Other transport (incl. aerospace)	0.10	0.10	0.157	3.1	3.1
21	Pharmaceuticals	0.10	0.00	0.130	2.6	0.0
27	Electrical equipment	0.10	0.10	0.126	2.5	2.5
20	Chemicals	0.10	0.10	0.125	2.5	2.5
24	Basic metals (incl. steel)	0.25	0.125	0.051	2.5	1.3
11	Beverages	0.10	0.10	0.063	1.3	1.3

Notes: $s_j = \varepsilon\,\tau_j\,x_j$ with $\varepsilon = 2.0$ and pass-through 1.0. Intensity is 2024 US goods exports over ONS ABS division turnover (Appendix A). The EPD steel rate is the parameterised half-relief of the 25 per cent rate; the auto rate is the in-quota 10 per cent.

of work; the EPD schedule 27,700 (±4,800; the quota expectations are 33,566 and 26,709). The age profile of the displaced skews old relative to the workforce—across draws, 18 ± 10 per cent are aged 55–64 and 23 ± 12 per cent are 55 or over—which is what gives the inactivity margin its potential bite (and Section 4.4 its scoped limitation).

4.2 Cushioning: the tax–benefit system absorbs wage cuts more than job loss

Table 2 makes the headline finding precise. Define the cushioning rate as one minus the ratio of the aggregate household disposable income loss to the aggregate gross earnings loss. Under the full schedule, the displacement margin turns a £1.70 billion gross earnings loss (seed-0 draw) into a £1.14 billion disposable income loss—a cushioning rate of 32.8 per cent—while the earnings-equivalent wage-cut margin turns £1.77 billion of gross earnings into a £1.04 billion disposable loss, a cushioning rate of 41.0 per cent. The same ordering holds under the EPD schedule (35.1 versus 40.8 per cent), so it is robust to the composition of the shock across sectors.

Table 2: Cushioning rates by tariff scenario and adjustment margin (seed 0)

Tariff scenario	Margin	Gross earnings loss (£bn/yr)	Net disposable loss (£bn/yr)	Cushioning rate (%)
Full tariffs	Displacement	1.70	1.14	32.8
Full tariffs	Wage cut	1.77	1.04	41.0
EPD	Displacement	0.91	0.59	35.1
EPD	Wage cut	1.39	0.82	40.8

Notes: cushioning rate = 1 − (net disposable loss)/(gross earnings loss), aggregated over all households. Displacement rows are the seed-0 draw (the wage-cut margin is deterministic); the displacement gross loss differs slightly from the expected £1.77bn/£1.39bn because a single draw realises the quota on discrete workers. The EPD displacement gross loss is smaller than the EPD wage-cut loss for the same reason compounded by which sectors' quotas round down at seed 0.

This result reverses the prior stated in Section 3.3, and rather than assert a mechanism I demonstrate it: Section 4.3 decomposes both cushioning rates into named tax and benefit components, counts which UC constraint binds on each displaced worker's family, and reruns the displacement scenario with the UC capital rules switched off. To preview the conclusion: the gap is carried by income-tax arithmetic (marginal versus average rates) reinforced by the gated extensive margin of UC—where the gates that actually bind in the model are modelled take-up and the joint partner-income taper, not the capital test. In the language of Atkin et al. (2025), the statutory generosity of out-of-work support is not its economic generosity once the gates bind on the population actually displaced. The Dolls et al. (2012) finding that European systems absorb unemployment shocks *more* than income shocks does not transfer to a UK manufacturing displacement on these data.

4.3 The mechanism, demonstrated

Table 3 decomposes each cushioning rate exactly. Because HBAI net income is an additive list of income, tax and benefit components, the cushioned share of the gross earnings loss splits without residual approximation into: foregone income tax, foregone employee National Insurance, additional Universal Credit, additional other benefits (JSA, ESA, housing benefit, tax credits, pension credit and the rest of the HBAI benefit list), and a remainder (zeroed employee pension contributions, statutory pay, council tax interactions).

Table 3: Component decomposition of the cushioning rate, full tariff schedule

Component (share of gross earnings loss, %)	Displacement (seed 0)	Wage cut
Income tax foregone	19.8	35.2
Employee National Insurance foregone	4.5	4.3
Universal Credit response	4.2	1.5
Other benefits (JSA, ESA, HB, tax credits, ...)	0.0	0.1
Other (pension contributions, statutory pay, ...)	4.3	0.0
Total cushioning rate	32.8	41.0

Notes: components of $1 - \Delta(\text{HBAI net income})/\Delta(\text{gross earnings})$, aggregated over all households; columns sum to the total by construction. Displacement is the seed-0 draw; the wage-cut margin is deterministic.

The decomposition overturns part of the story I would otherwise have told, and I report what it shows rather than what the prior predicted. The 8.2-percentage-point gap is more than fully accounted for by income tax: the wage-cut margin recovers 35.2 per cent of the gross loss in foregone income tax against 19.8 per cent under displacement, a 15.4-point difference that alone exceeds the whole gap. The reason is marginal-versus-average arithmetic, not means testing: a proportional wage cut shaves the *top slice* of every affected worker's earnings, relieved at marginal rates, whereas displacement removes the *entire* earnings stream—including the slice covered by the personal allowance and taxed at zero—so the tax system hands back only the average rate. National Insurance contributes equally on both margins (4.5 versus 4.3 per cent; NI's flatter schedule means its marginal and average rates differ less). Universal Credit runs the *other* way—it cushions displacement more (4.2 versus 1.5 per cent), as the naive replacement-rate intuition says it should—but far too weakly to offset the tax asymmetry, and the other-benefits row contributes nothing at all under displacement: in particular, new-style contribution-based

JSA pays zero to the displaced in the model, a modelling scope limit (JSA claims are not activated by the displacement transition) that should be read alongside the UC findings below.

Why is the UC response to a mass displacement only 4 pence per pound of earnings lost? Counting displaced workers' benefit units post-shock (seed 0): 33 per cent receive positive UC, averaging £7,600 a year conditional on receipt; 67 per cent receive none (across seeds 0–4 the zero-UC share is 71 ± 16 per cent). Decomposing the zero-UC share by the constraint that binds: 55 per cent of all displaced workers are in families modelled as not taking up UC (PolicyEngine's stochastic take-up draw, held fixed from the baseline), 12 per cent are in families whose joint means test tapers the maximum award to zero against a partner's earnings and unearned income (18 ± 9 per cent across seeds), and—contrary to the mechanism the working-paper draft asserted—0 per cent are excluded by the £16,000 capital limit. The capital result is a data limitation, not a finding about the rules: the plain FRS 2024–25 dataset used here carries no household capital (the UC-countable capital variables are zero throughout), so the capital test *cannot* bind in the model, and this paper cannot test how many displaced workers it would disqualify in reality. A counterfactual rerun makes the same point mechanically: neutralising the UC capital rules (raising the £16,000 limit and the £6,000 tariff-income threshold by five orders of magnitude, taper and work allowances unchanged) leaves the displacement cushioning rate identical to four decimal places (32.83 per cent), where the asserted mechanism predicted it would rise toward the wage-cut margin's 41 per cent. The take-up gate, by contrast, is quantitatively live: forcing 100 per cent UC take-up in both the baseline and shocked simulations (the plain FRS dataset's modelled take-up flags imply 55 per cent of individuals are in benunits that would claim) more than doubles the UC response to displacement (£0.16 versus £0.07 billion a year) and raises the displacement cushioning rate from 32.9 ± 2.6 to 38.6 ± 2.5 per cent over 10 common draws—closing about seventy per cent of the gap to the wage-cut margin's 41 per cent (Appendix E).

The corrected reading of the headline result is therefore: the tax–benefit system cushions wage cuts more than job losses *first* because income tax relieves marginal earnings at marginal rates but lost jobs only at average rates, and *second* because the extensive margin of UC is gated—by modelled take-up and by the joint partner-income taper on these data; plausibly also by the capital test in reality, but that channel is unobservable in the plain FRS and contributes nothing here. The qualitative claim that survives intact is that out-of-work support reaches only a third of the displaced while in-work tax relief reaches every wage-cut worker automatically; the specific attribution to the capital means test does not.

4.4 Distributional incidence by adjustment margin

Figure 1 shows the mean change in household disposable income by equivalised income decile under the three margins, full schedule, Monte Carlo means over the 50 draws with ±1 SD bands. Two profiles emerge. The displacement margin is top-half concentrated: mean losses are largest in deciles 7–10 (£71–174 per person per year, with draw-to-draw SDs of £43–135 in those deciles—individual draws are lumpy, reflecting which households happen to contain the drawn workers) and small in the bottom four deciles (under £20). The wage-cut margin is smooth and rises monotonically with income, from £6 in the bottom decile to £116 at the top: exposed manufacturing earnings sit disproportionately in the upper-middle of the distribution, and proportional cuts scale with earnings.

This decile profile yields a named finding: *the tariff shock is regionally, not vertically, concentrated.* Displacement losses land hardest on deciles 7–10—manufacturing employees are mostly

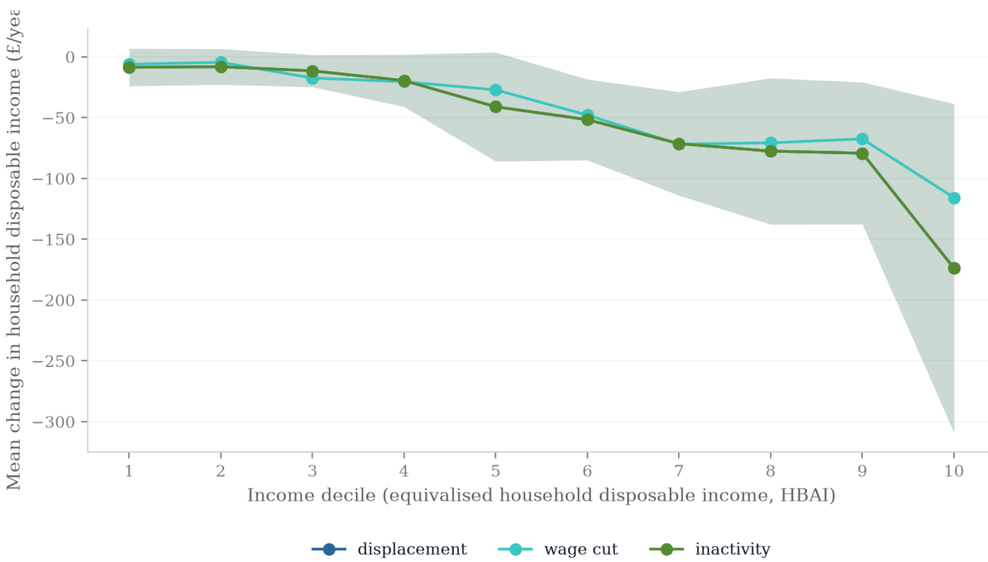

Figure 1: Mean change in household disposable income by equivalised income decile under the three adjustment margins, full tariff schedule. Lines are Monte Carlo means over 50 displacement draws; shaded bands are ±1 draw-to-draw SD (the wage-cut margin is deterministic). The inactivity margin coincides with displacement in the current model version (see text).

not poor—so the aggregate inequality and poverty statistics barely move. The Gini coefficient of equivalised disposable income rises by 0.00022 ± 0.00017 under full displacement (from a baseline of 0.309), and actually *falls* very slightly (-0.0001) under the wage-cut margin, since proportional cuts to above-median earnings are mildly equalising. Relative BHC poverty rises by 0.055 ± 0.021 percentage points under full displacement—roughly 38,000 people—and by essentially zero under the wage-cut margin: a several-per-cent wage cut pushes almost no exposed household across the poverty line, while a job loss pushes some. Absolute BHC poverty against a fixed baseline line (computed manually at seed 0; Appendix C) moves nearly identically (+0.058 points full displacement, +0.029 EPD, zero under wage cuts), confirming that the relative-line result is not an artefact of a shock-shifted median.

The near-null national headcount conceals an intensive margin that the poverty *gap* makes visible. The aggregate relative BHC poverty gap—the weighted sum of household shortfalls below the line—rises by £0.19 billion a year ± 0.05 under full-schedule displacement (£0.14 billion ± 0.06 under the EPD) on a national baseline gap of £36.3 billion: a 0.5 per cent movement, invisible nationally, and only £0.014 billion (deterministic) under the wage-cut margin. But essentially the entire gap increase is borne inside *affected* households (those containing a displaced worker; 20 draws, seeds 0–19): their own baseline poverty gap of £0.01 billion rises twenty-fold to £0.19 billion post-shock, and the relative BHC poverty rate among the roughly 94,000 people in those households jumps from 4 ± 6 to 43 ± 13 per cent—a rise of 39 ± 12 percentage points (34 ± 13 points under the EPD). Among the 4.6 million people in wage-cut-affected households, by contrast, the poverty rate does not move at all. The correct reading of the "+0.055 points" headline is therefore not that the shock is poverty-irrelevant, but that a full-year displacement is close to poverty-*certain* for the small population it hits, while remaining invisible in national statistics. The distributional action, such as it is, therefore runs along the margin (who loses work versus who loses wages) and along geography (Section 4.8)—not down the income distribution.

The inactivity margin comes with a scoped limitation that I state plainly rather than bury: in policyengine-uk 2.89.2 the transition to economic inactivity is applied and verified (displaced over-50s are set to inactive rather than unemployed, and the run hard-errors if the status change fails to apply), but absent modelling of the UC health element and of conditionality, an inactive claimant's computed entitlement is indistinguishable from an unemployed one's. Every inactivity-margin result in this paper is therefore numerically identical to its displacement counterpart. The margin is retained in the scenario grid because the machinery is in place and because UK history says it matters (Beatty et al., 2007; Beatty and Fothergill, 2017); quantifying its distinct fiscal cost—the health-element premium and the absence of work-search conditionality—is the first item of future work (Section 6).

4.5 Who bears it: gender, age and household type

Tables 4 and 5 report who the displaced are and how much of their loss the state absorbs, over 20 Monte Carlo draws (seeds 0–19; full schedule). Group cushioning rates attribute each household's disposable-income change to its affected workers in proportion to their gross earnings losses, so the rows aggregate exactly to the economy-wide rates of Table 2.

Three patterns are worth naming, with the standing Monte Carlo discipline that group-level draws are lumpy. First, gender: 73 ± 9 per cent of the displaced are men—exposed manufacturing is male-dominated—but the system cushions displaced women *less* (28 ± 7 against 35 ± 5 per cent), a one-sided contrast in almost every draw even though the means sit within one joint SD. Under the wage-cut margin the gender gap in cushioning nearly closes (40.4 versus 41.2 per cent): the asymmetry is a property of the gated out-of-work system, not of the tax schedule. This is the tax–benefit-system counterpart of the De Lyon and Pessoa (2021) finding that UK women bore the China shock disproportionately through lost employment years: here, when women do lose manufacturing jobs, the state also replaces less of the loss. Second, age: displacement is concentrated in prime age (30 and 25 per cent of the displaced are 35–44 and 45–54 respectively), and cushioning is weakest at the two ends—29 ± 4 per cent for under-25s and 27 ± 12 per cent for the 65-plus—the groups least likely to clear UC's gates. Third, household type, which is where the mechanism section's partner-income taper becomes visible in the incidence: half the displaced (50 ± 14 per cent) live in dual-earner couples, and their cushioning rate (32.5 ± 4.9 per cent) sits below single-earner couples' (36.4 ± 7.6)—directionally the taper gate at work (a working partner's earnings exhaust the UC award), though the two means sit within one SD of each other and I flag rather than lean on the pairwise ordering. The cleaner statement is again the cross-margin one: under wage cuts, where UC is largely irrelevant, dual- and single-earner couples are cushioned almost identically (42.0 versus 43.1 per cent); the differential opens only on the displacement margin, where the joint means test operates.

4.6 Fiscal incidence

Table 6 reports the annual Exchequer cost—foregone tax plus additional benefit spending—by scenario, and Figure 2 shows the full draw distribution. Under the full schedule, displacement costs £0.85 billion ± 0.30 (mean ± SD over 50 draws) and the wage-cut margin £0.98 billion; under the EPD, £0.66 billion ± 0.33 and £0.77 billion. The wage-cut margin costing *more* than displacement is the fiscal mirror of the cushioning result: the Exchequer absorbs wage cuts automatically at marginal income-tax rates for every affected worker, whereas a displacement

Table 4: Displacement incidence by gender and age band, full tariff schedule

Group	Displacement (20 draws) Share of displaced (%)	Displacement (20 draws) Mean income change per displaced (£/yr)	Displacement (20 draws) Cushioning rate (%)	Wage cut Cushioning rate (%)
Men	73 ± 9	$-37{,}900 \pm 8{,}800$	35 ± 5	41.2
Women	27 ± 9	$-28{,}600 \pm 9{,}800$	28 ± 7	40.4
16–24	7 ± 4	$-27{,}200 \pm 9{,}800$	29 ± 4	29.6
25–34	19 ± 8	$-26{,}800 \pm 7{,}500$	30 ± 8	37.8
35–44	30 ± 12	$-43{,}100 \pm 22{,}400$	33 ± 7	42.9
45–54	25 ± 7	$-34{,}000 \pm 8{,}700$	34 ± 6	45.2
55–64	17 ± 10	$-31{,}100 \pm 12{,}000$	33 ± 10	39.1
65+	7 ± 6	$-41{,}200 \pm 66{,}000$	27 ± 12	34.8

Notes: Monte Carlo means ± SD over 20 displacement draws. Income changes are the household HBAI net-income change attributed to affected workers in proportion to their gross earnings losses, per displaced worker per year; group cushioning is 1− attributed net loss / gross earnings loss. The wage-cut column is the deterministic run's group cushioning rate (its per-worker income changes, £400–900 a year, are on a different scale and omitted).

Table 5: Displacement incidence by household type, full tariff schedule

Household type	Displacement (20 draws) Share of displaced (%)	Displacement (20 draws) Mean income change per displaced (£/yr)	Displacement (20 draws) Cushioning rate (%)	Wage cut Cushioning rate (%)
Single, no children	24 ± 9	$-28{,}700 \pm 6{,}100$	33 ± 5	36.9
Single, with children	6 ± 3	$-30{,}900 \pm 20{,}600$	33 ± 14	44.0
Couple, no children, single earner	12 ± 8	$-36{,}300 \pm 23{,}800$	34 ± 10	40.2
Couple, no children, dual earner	24 ± 11	$-35{,}200 \pm 9{,}500$	29 ± 6	38.5
Couple, with children, single earner	10 ± 5	$-49{,}000 \pm 43{,}900$	34 ± 10	45.8
Couple, with children, dual earner	26 ± 13	$-35{,}500 \pm 9{,}200$	34 ± 6	45.2
All couples, single earner	21 ± 11	$-43{,}500 \pm 27{,}000$	36 ± 8	43.1
All couples, dual earner	50 ± 14	$-34{,}900 \pm 6{,}200$	33 ± 5	42.0

Notes: as Table 4. Household type is the displaced worker's benunit family type at baseline; earner counts are baseline counts of benunit adults with positive employment income. The bottom panel pools couples across children status.

destroys tax liability only at average rates and—with two-thirds of displaced families receiving no UC (Section 4.3)—triggers comparatively little benefit spending. That said, MC discipline applies: the displacement–wage-cut gap of £0.13 billion (full schedule; £0.11 billion EPD) is smaller than the £0.30 billion draw-to-draw standard deviation, and the deterministic wage-cut cost exceeds the displacement draw in 37 of 50 draws under each schedule—suggestive and mechanism-consistent, but not sharply identified by the draws alone; the cushioning rates of Table 2, which compare aggregates rather than draws, carry the identification. Orderings that clearly survive the SD: every scenario costs the Exchequer strictly more than zero. The full schedule costs more than the EPD within each margin in expectation, but this too is a paired-draws statement: the full−EPD difference on common seeds averages £0.19 billion ± 0.38 and is positive in 36 of 50 paired draws—directionally consistent, though individual draws can reverse it.

Table 6: Annual Exchequer cost by scenario (2026, £bn)

	Displacement	Wage cut	Inactivity
Full tariffs	0.85 ± 0.30	0.98	0.85 ± 0.30
EPD	0.66 ± 0.33	0.77	0.66 ± 0.33

Notes: means ± SD over 50 Monte Carlo displacement draws; the wage-cut margin is deterministic. Inactivity equals displacement in the current model version (see Section 4.4). Fiscal policy held fixed; no revenue recycling (Ignatenko et al., 2025).

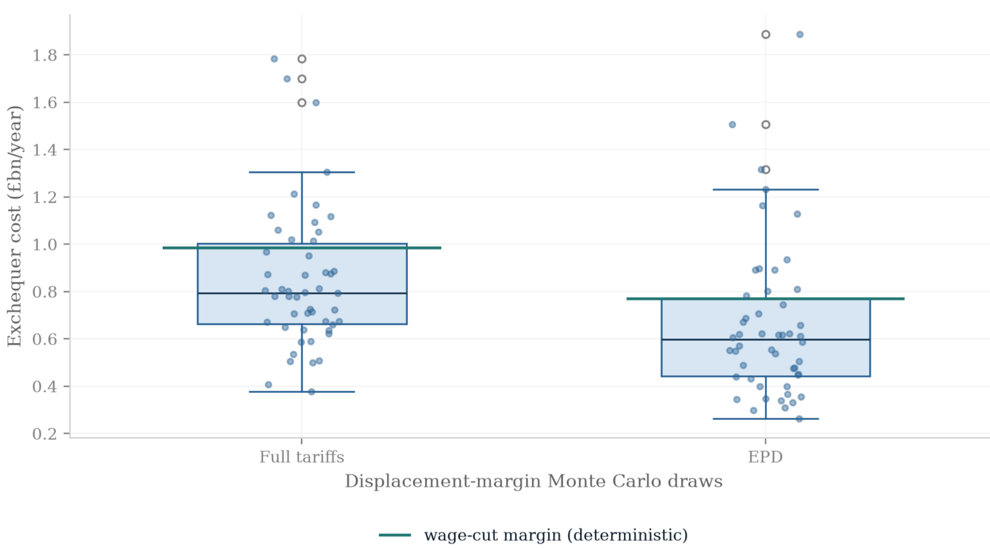

Figure 2: Draw distribution of the annual Exchequer cost on the displacement margin: box plots and individual draws over the 50 Monte Carlo displacement draws, full tariffs and EPD, with the deterministic wage-cut cost overlaid as a horizontal line. The wage-cut cost sits above the displacement draw in 37 of 50 draws under each schedule, but inside the interquartile whisker range—the ordering is suggestive, not sharply identified by the draws.

The displacement costs above annualise a full-year out-of-work spell; they scale roughly pro rata with the expected duration. Rerunning displacement with displaced workers keeping half a year's earnings (a six-month expected spell, evaluated in-model so taxes and benefits respond to the actual annual income) cuts the Exchequer cost from £0.70 to £0.41 billion over 10 common draws—slightly *more* than half, because the retained half-year of earnings is taxed at rates above the displaced worker's average rate—and lifts the cushioning rate from 33 to 42 per cent, while the poverty effect nearly vanishes (+0.013 against +0.056 percentage points): half a year of earnings keeps most displaced households above the line (Appendix E).

Whichever margin, the fiscal incidence dwarfs the poverty incidence. Set against a poverty rise of at most six-hundredths of a percentage point, an annual Exchequer cost of £0.7–1.0 billion says the 2025 tariff reaches the UK household sector primarily as a public-finance problem, mediated by the automatic stabilisers, rather than as a poverty problem.

4.7 The EPD counterfactual

Table 7 and Figure 3 price the Economic Prosperity Deal as the difference between the two tariff scenarios run through identical machinery. On the displacement margin the deal averts about 6,900 job losses in quota expectation (33,566 versus 26,709; the Monte Carlo means over 50 draws are 34,200 versus 27,700), £0.38 billion of gross annual earnings loss (£1.77 versus £1.39 billion), and roughly £0.2 billion of annual Exchequer cost (£0.19 billion ± 0.38 mean paired-draw difference on the displacement margin, positive in 36 of 50 paired draws; £0.22 billion, deterministic, on the wage-cut margin). Three sectoral clauses do all the work. The auto rate cut from 25 to the in-quota 10 per cent more than halves the largest single-sector shock, and because the 100,000-vehicle quota approximately equals 2024 US export volumes, the in-quota rate is the operative one. The pharmaceutical exemption removes the SIC 21 employment shock entirely—by construction, since the exemption's price was paid elsewhere (Section 5.2). Steel relief is modelled as partial (half the 25 per cent rate), and carries the standing caveat that with roughly 80 per cent of UK steel exports going to the EU, the US-tariff cell understates the sector's true trade-policy exposure, which is dominated by EU/UK safeguards.

Table 7: What the EPD averted: full tariffs versus EPD, annual

	Full tariffs	EPD	Averted by EPD
Displaced workers (expectation)	33,566	26,709	6,857
Displaced workers (MC mean ± SD)	34,200 ± 3,600	27,700 ± 4,800	6,500
Gross earnings loss (£bn)	1.77	1.39	0.38
Exchequer cost, displacement margin (£bn)	0.85 ± 0.30	0.66 ± 0.33	0.19 ± 0.38
Exchequer cost, wage-cut margin (£bn)	0.98	0.77	0.22
Relative BHC poverty change (ppt, displacement)	+0.055 ± 0.021	+0.042 ± 0.022	0.013

Notes: displacement rows are means over 50 Monte Carlo draws (± SD); averted Exchequer costs are mean paired-draw differences on common seeds (± SD of the paired difference; positive in 36 of 50 draws). The wage-cut margin is deterministic.

4.8 Regional incidence

The regional profile is where the shock's concentration actually lives, and it moves with the margin (Figure 4; the ITL1 regional map is Figure 5 and the full three-scenario table is Table 10,

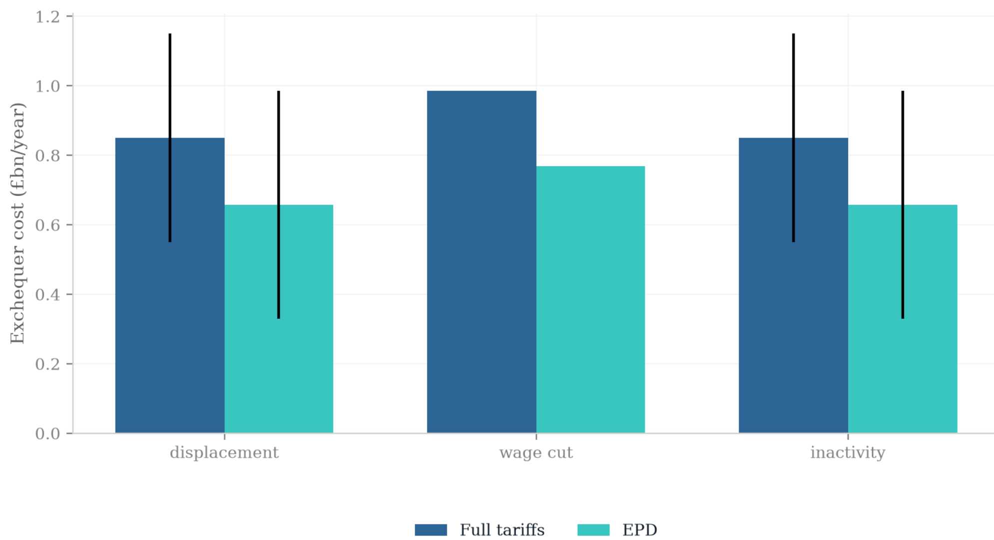

Figure 3: Annual Exchequer cost, full tariffs versus EPD, by adjustment margin. Bars are means over 50 draws; error bars are draw-to-draw standard deviations (wage cut is deterministic).

both in Appendix F; all regional displacement numbers are Monte Carlo means ± SD over the 50 draws). Under full-schedule displacement the largest mean regional losses are Wales (£108 ± 234 per person per year—the enormous SD reflects a single high-weight Welsh record that dominates the draws it enters), the East of England (£74 ± 76), the East Midlands (£72 ± 73, reflecting machinery, transport-equipment and food-manufacturing employment), the West Midlands (£64 ± 56) and Scotland (£62 ± 50, with beverages, i.e. Scotch whisky at a 6.3 per cent US-export intensity, prominent). No pairwise regional ordering survives one draw-to-draw SD, and I do not assert one. Under the deterministic wage-cut margin the same aggregate loss spreads smoothly: Wales (£65), the East of England (£61) and the West Midlands (£55) lead, because the proportional cut weights regions by their exposed *wage bills* rather than by where the discrete displacement draw lands. Under EPD displacement the mean profile is led by Wales (£75 ± 181), the East Midlands (£70 ± 93) and Northern Ireland (£63 ± 52); Yorkshire and the Humber, the steel region for which the deal's conditional half-relief is the only mitigation, averages £34 ± 37—an earlier draft's single-draw figure of £121 for Yorkshire illustrates exactly how far one draw can sit from the mean. The concentration signature is therefore a statement about *within-draw* geography: in any single draw the displacement losses pile up in a handful of regions—which is why the region-level SDs are comparable to or larger than the means—while wage cuts spread thinly everywhere. That is the margin's geographic signature, and it is the regional counterpart of the Gini result: a shock that is invisible in national inequality statistics can still be a £100–200-per-head event in whichever regions a given realisation hits.

The regional concentration sharpens further at constituency level. Projecting the two displacement scenarios onto the 650 Westminster parliamentary constituencies with PolicyEngine's calibrated 650×53,508 constituency weight matrix (Figure 4) shows the full-schedule loss concentrating in Scottish and Northern Irish seats. These constituency results are a single-draw (seed-0), imputation-based illustration—unlike the Monte Carlo means above, the rankings would reshuffle under other draws: in that draw the ten hardest-hit constituencies under full-tariff displacement are West Dunbartonshire ($-£390$ per person per year), Glasgow South West ($-£362$), Dumfries and Galloway ($-£322$), Belfast East ($-£301$), Lothian East ($-£283$), Belfast South and Mid Down ($-£239$), North East Fife ($-£233$), Paisley and Renfrewshire South ($-£229$), Caithness, Sutherland and Easter Ross ($-£227$) and Paisley and Renfrewshire North ($-£223$)—eight Scottish seats, consistent with the beverages division's outsize US-export intensity (Scotch whisky), plus the two most manufacturing-intensive Belfast seats. Under the EPD (same seed-0 draw) the map empties across Great Britain but Northern Ireland's residual exposure dominates: nine of the ten hardest-hit seats are Northern Irish, led by Belfast East ($-£498$), Belfast South and Mid Down ($-£351$) and Belfast West ($-£302$), with Sheffield Central ($-£103$)—a steel seat under the deal's half-relief—the only Great British entrant. West Midlands automotive seats (Solihull West and Shirley, Meriden and Solihull East) sit in the hardest-hit quartile under the full schedule but nowhere near the top ten, because the EPD-relieved auto rate and the discrete displacement draw both dilute their exposure. One methodological caveat governs these numbers, mirroring the sister study's constituency analysis: the weight matrix indexes the *enhanced* FRS 2023–24 dataset, which carries no industry codes, so SIC division is imputed from the plain-FRS 2024–25 weighted distribution within age-band $\times$ gender $\times$ region $\times$ earnings-decile cells (99.4 per cent weighted coverage from the full four-way cell, seed 0). Constituency variation therefore reflects demographic and earnings composition under a single seed-0 draw and model-calibrated weights, not observed local industry mix, and is computed on a different dataset and period (2025) from the headline runs—indicative of relative, not absolute, local impacts.

4.9 Validation against the 2025–26 outturn

The design lets the realised data discipline the one free parameter. ONS monthly trade data show UK goods exports to the US falling 24.7 per cent in April 2025, the first full tariff month; against the 12.8 per cent trade-weighted average tariff implied by the primitives in Table 1, this pins the export-demand elasticity at 1.93, which I round to 2.0. The calibration is deliberately an *outturn* anchor rather than a literature import: it reflects the actual short-run US demand response to this schedule, including anticipation and front-running effects that average out over the calendar year. The sectoral outturn is consistent with the derived shock ranking: SMMT monthly data show US-bound vehicle exports collapsing from 7,077 units in April to 4,223 in May 2025 before recovering after the EPD's implementation, exactly the autos-led profile Table 1 derives. Which *adjustment margin* the data chose is harder to read this early: BICS waves through 2025–26 show exposed manufacturers reporting price and margin adjustments more often than redundancies, which favours the wage-cut family as the short-run description—and therefore, by Section 4.2, the higher-cushioning, higher-Exchequer-cost, lower-poverty realisation of the shock.

5 Policy

5.1 What the Economic Prosperity Deal was worth

Priced as the difference between the two tariff scenarios run through identical machinery (Section 4.7), the EPD is worth, per year on the displacement margin: about 6,900 manufacturing jobs in expectation, £0.38 billion of gross earnings, and roughly £0.2 billion to the Exchequer (£0.19 billion $\pm$ 0.38 mean paired-draw difference on the displacement margin, positive in 36 of 50 draws; £0.22 billion, deterministic, on the wage-cut margin). Relative BHC poverty impacts

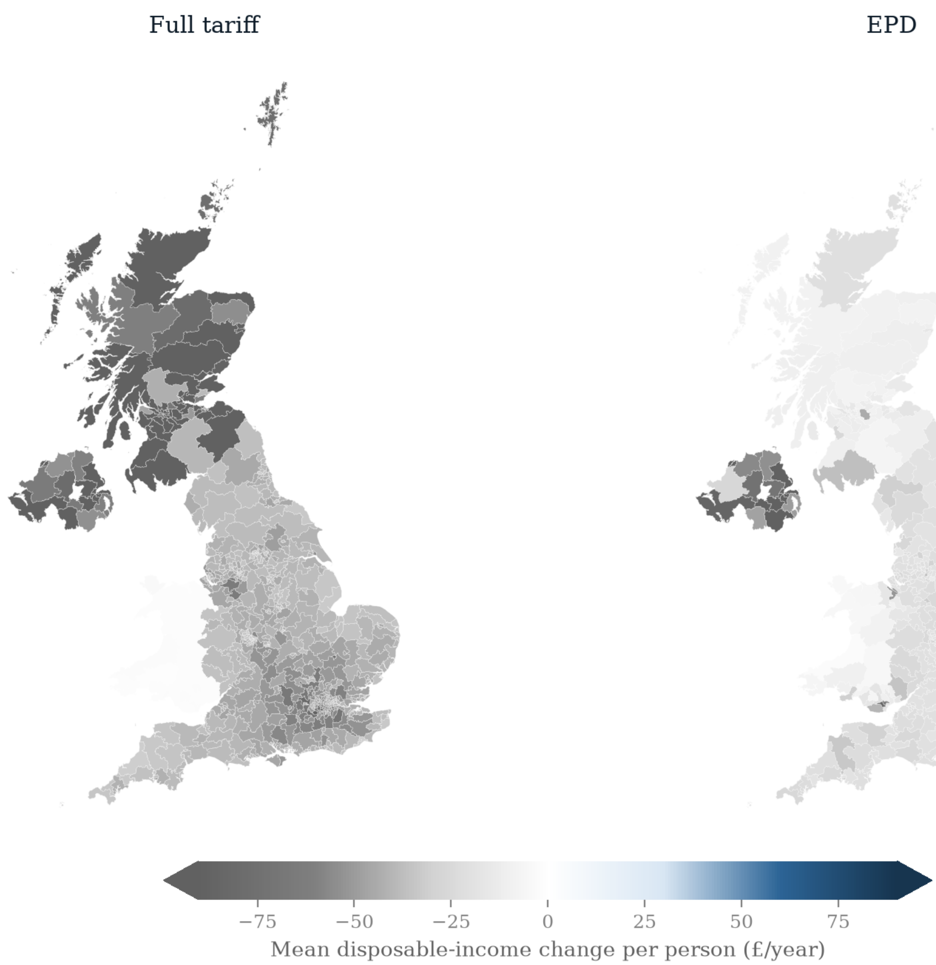

Figure 4: Constituency choropleth maps of the mean disposable-income change per person per year (£, seed-0 draw) across the 650 Westminster parliamentary constituencies: full-tariff displacement (left) and EPD displacement (right), on a shared diverging scale (losses in grey), clipped at the 95th percentile of the absolute change (caps show clipping). Computed on the enhanced FRS 2023–24 dataset with PolicyEngine's calibrated constituency weight matrix and imputed SIC divisions, period 2025; constituency variation reflects demographic and earnings composition under a single displacement draw, not observed local industry mix, and is indicative of relative rather than absolute local impacts. The ITL1 regional map is Figure 5 in Appendix F.

fall from $+0.055 \pm 0.021$ to $+0.042 \pm 0.022$ percentage points under displacement. These are modest absolute numbers—the direct earnings channel of the full schedule was itself only 0.06 per cent of GDP—but they are not evenly spread. The deal's value is a bundle of three sectoral clauses with three distinct geographies. The auto quota (100,000 vehicles at 10 per cent, approximately the 2024 US export volume, so the in-quota rate is the effective one) removes the single largest derived shock and is worth the most to the West Midlands and the wider vehicle supply chain. The pharmaceutical exemption removes the SIC 21 employment shock entirely, protecting a high-wage, geographically dispersed sector—but its price was paid through a different fiscal door (below). Steel relief, modelled at half the 25 per cent rate, is the weakest clause: it was conditional and slow to implement, it leaves Yorkshire and the Humber still exposed under EPD displacement (a Monte Carlo mean loss of £34 ± 37 per person per year; single draws have put it as high as £121), and in any case the steel sector's binding trade-policy exposure is EU/UK safeguards, not the US tariff. What the deal did *not* do is also visible in Table 1: machinery, electronics, aerospace and chemicals—which together account for more derived earnings loss than autos—remain at the 10 per cent baseline, which is why the EPD removes only about a fifth of the gross earnings shock.

5.2 The pharmaceutical channel

The pharmaceutical exemption deserves its own accounting because its incidence is fiscal and price-side, not an employment shock—the cleanest illustration in this paper of statutory versus economic incidence (Atkin et al., 2025). Statutorily, the December 2025 side agreement zero-rates UK pharmaceutical exports to the US for three years. Economically, the exemption was purchased through the NHS: the VPAG rebate rate on branded medicine sales was cut from 22.5 to 14.5 per cent, alongside NHS pricing concessions (ABPI, 2025). With UK branded-medicine sales subject to VPAG in the tens of billions, an eight-percentage-point rebate cut plausibly costs the health budget an amount of the same order as the entire Exchequer cost of the tariff shock computed in Section 4.6—a transfer from the NHS to the pharmaceutical sector's US price exposure. This paper's employment-channel machinery correctly assigns pharma a zero shock under the EPD, but the reader should not conclude the exemption was free: its cost simply does not pass through household earnings, and therefore does not appear in a tax–benefit microsimulation. A full welfare accounting of the EPD would set the £0.2 billion annual Exchequer saving found here against the VPAG concession on the other side of the public balance sheet.

5.3 Cushioning reforms

The cushioning result of Section 4.2 reframes the standard policy menu. The decomposition of Section 4.3 shows the binding constraint on cushioning displaced workers is not the generosity of out-of-work support but the gates on it: on these data, modelled take-up and the joint partner-income taper leave two-thirds of displaced workers' families with no UC at all, so reforms that raise UC rates or cut the taper do little for the displaced worker whose family never enters an award calculation. Three directions follow, all simulable in this framework. First, gate easements for the involuntarily displaced: automatic enrolment (or an assumed-take-up presumption) after redundancy, easing the joint partner-income assessment, and disregarding savings for a fixed period—the last consistent with, though not testable on, FRS data that carry no household capital—would extend to job losers part of the automatic cushioning that wage-cut workers already enjoy. Even full UC coverage would not close the 33-versus-41 per cent gap by itself, however: most of the gap is income-tax arithmetic (marginal versus average rates), not benefit design. Second, extending the duration or level of contribution-based new-style JSA, the instrument that reaches displaced workers regardless of take-up frictions and partner income—a UK analogue of earnings-related unemployment insurance. Third, wage insurance in the spirit of Hyman et al. (2024), which subsidises re-employment at lower wages: in the language of this paper it converts a displacement-margin realisation of the shock into a wage-cut-margin one, and Section 4.2 shows the UK system is already better at absorbing exactly that margin—so wage insurance and the existing UC taper are complements, not substitutes. Throughout, the revenue-recycling caveat of Ignatenko et al. (2025) applies: this paper prices reforms in a first-round world with fiscal policy otherwise fixed, and financing choices can alter even the sign of aggregate employment effects.

6 Discussion and conclusion

This paper asked who bears the 2025 US tariffs on UK goods after the tax–benefit system responds, and answered by deriving the shock from primitives—the tariff schedule, HMRC trade-by-SIC exposure, and an outturn-anchored export-demand elasticity—and passing it through PolicyEngine UK under three adjustment margins holding the aggregate earnings loss fixed. Three conclusions stand.

First, the adjustment margin, not the tariff rate alone, determines what the shock *is* when it reaches households—and the direction of that determination reverses the naive prior. The UK tax–benefit system cushions an in-work wage cut (41 per cent) more than an earnings-equivalent wave of job losses (33 per cent), and the decomposition of Section 4.3 shows why: income tax relieves a wage cut at marginal rates but a lost job only at average rates (35 versus 20 pence per pound of gross loss), and Universal Credit—though it responds more to displacement, as the naive intuition predicts—reaches only a third of displaced workers' families, gated in the model by take-up and the joint partner-income taper rather than by the capital limit, which the plain FRS data (carrying no household capital) cannot exercise. Statutory generosity toward the unemployed is not economic generosity once the gates bind on the population actually displaced. The fiscal ordering is the mirror image, though it is suggestive rather than sharply identified: the deterministic wage-cut margin costs the Exchequer £0.98 billion a year under the full schedule against a displacement mean of £0.85 billion $\pm$ 0.30 across 50 draws—an ordering that holds in 37 of 50 draws but whose £0.13 billion gap sits inside one draw-to-draw SD—while only the displacement margin moves poverty at all ($+0.055 \pm 0.021$ percentage points versus approximately zero).

Second, on the displacement margin the shock is regionally, not vertically, concentrated. Displaced manufacturing earnings sit in deciles 7–10, so national inequality and poverty statistics barely register a shock whose Monte Carlo mean regional losses reach £60–110 per person per year in the hardest-hit regions—and, within any single realisation, £100–200 per head in whichever handful of regions the displacement draw lands on. Trade-shock incidence in the UK is a geography problem wearing the clothes of a distribution problem—consistent with the China-shock literature's regional persistence findings (Autor et al., 2021; Dix-Carneiro and Kovak, 2017) but established here at the household, rules-based level.

Third, the Economic Prosperity Deal was worth about 6,900 jobs (in expectation), £0.38 billion of gross earnings and roughly £0.2 billion a year to the Exchequer on the direct channel (a mean paired-draw difference of £0.19 billion ± 0.38, positive in 36 of 50 draws)—real but modest, concentrated in autos, with the pharmaceutical clause's true cost routed through the NHS rather than the labour market and the steel clause largely beside the sector's EU-dominated exposure.

Limitations. Five are worth stating plainly. (1) The inactivity margin is currently indistinguishable from displacement: policyengine-uk 2.89.2 applies and verifies the status transition, but without UC health-element and conditionality modelling an inactive claimant's computed entitlement equals an unemployed one's. The Beatty–Fothergill margin is therefore scoped, not quantified; the UC-health-element extension is the first item of future work—and an urgent one, since incapacity claims still respond to local labour-market slack (Roberts and Taylor, 2022) and the health-related caseload was already at a post-pandemic high when the tariffs arrived (IFS, 2024). (2) Monte Carlo dispersion is large relative to the means (£0.30 billion SD against £0.7–0.9 billion means over 50 draws), so only orderings that survive it are asserted; the displacement–wage-cut fiscal gap in particular (holding in 37 of 50 draws) is identified by the aggregate cushioning accounting rather than by the draws. (3) Absolute BHC poverty against a fixed baseline line is computed manually outside the model, which lacks an `in_absolute_poverty_bhc` variable in this version. (4) The analysis covers the direct exposed-sector earnings channel only: no supply-chain propagation—which the production-network literature finds can multiply direct effects severalfold (Barrot and Sauvagnat, 2016; Acemoglu et al., 2016)—no local-multiplier ripple effects, which US and German estimates put at 0.4–1.6 additional local jobs per lost tradable job (Moretti, 2010; Gathmann et al., 2020), no general-equilibrium wage responses, no exchange-rate offset (Corsetti et al., 2025), no retaliation, and fiscal policy held fixed—Ignatenko et al. (2025) show revenue recycling can flip aggregate employment signs, so the fiscal numbers are first-round incidence, not equilibrium outcomes. The 0.06-per-cent-of-GDP gross shock should be read as a lower bound on the all-channel effect the OBR puts at 0.3–0.6 per cent. (5) The trade numerator is on the HMRC RTS/OTS customs basis (£55.6 billion of 2024 US exports, £52.5 billion mapped to SIC divisions), which sits below the ONS balance-of-payments basis (about £59.3 billion); intensities inherit that coverage gap.

Further work. Beyond the health-element extension, three directions. A *measured-margin* family calibrated to the accumulating 2025–26 outturn (ONS trade, BICS, SMMT), replacing the stylised poles with the displacement/wage-cut mix the data actually chose, in the spirit of Rud et al. (2022)'s decomposition. Sharpening the constituency geography of Section 4.8—now mapped across all 650 Westminster seats, but on imputed industry codes and a single draw—with observed local industry structure (BRES employment by SIC) replacing the demographic imputation. And dynamic transition paths—re-employment hazards, scarring, and the multi-year fiscal tail that Dix-Carneiro (2014) and Beatty and Fothergill (2017) both imply a one-year snapshot understates.

The bottom line inverts where the paper began. The 2025 tariff reaches UK households mainly as a public-finance event, not a poverty event; and the arm of the welfare state that absorbs it best is not the safety net for the jobless but the automatic machinery that relieves every pound of in-work earnings at the marginal tax rate, with no claim to make and no gate to pass.

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A Data appendix: the trade-by-SIC build

The US-export intensity table x_j (Table 1) is built by `analysis/build_trade_by_sic.py` from public sources, with all raw API pulls cached in the repository.

Numerator. UK goods exports to the United States, calendar 2024, in pounds, from the HMRC `uktradeinfo` OData API—the machine-readable publication of HMRC Overseas Trade Statistics (OTS) and Regional Trade Statistics (RTS). The RTS endpoint is queried for CountryId 400 (United States), non-EU export flows, months 202401–202412, summed over all Government Office Regions, keyed by 2-digit SITC (rev. 4) chapter. Nine chapters that straddle SIC divisions (SITC 33, 66, 68, 71, 77, 78, 79, 87, 89) are refetched from the OTS endpoint at 3-digit SITC and split—for example, 714 aero engines map to SIC 30 (other transport) rather than SIC 28 (machinery)—with the build asserting that each OTS split reproduces its RTS chapter total to within rounding.

Crosswalk. SITC (rev. 4) to SIC 2007 division, coded explicitly in the build script. It is the standard manufacturing concordance plus documented judgement calls: SITC 681 (silver and platinum) and 896 (works of art) are excluded as London bullion- and art-market re-exports rather than UK production employment; 891 arms and ammunition maps to SIC 25; 872 medical instruments to SIC 32 with 871/873/874 to SIC 26; 776 semiconductors to SIC 26 with the rest of chapter 77 to SIC 27; 333 crude oil is dropped (extraction, negligible FRS employment) while 334/335 refined products map to SIC 19. Excluded chapters (91, 93, 96, 97 and non-goods special transactions) total under £0.3 billion.

Validation. The 2024 all-commodity total on the RTS/OTS customs basis is £55.6 billion, of which £52.5 billion maps to SIC divisions after the documented exclusions; the ONS balance-of-payments basis is about £59.3 billion, the difference being BoP coverage and valuation adjustments. Sector cross-checks pass: SITC 78 road vehicles £9.0 billion against SMMT's £8–9 billion; SITC 54 medicinal and pharmaceutical products £6.0 billion against ABPI's £6.6 billion (BoP basis). The US share of UK goods exports implied by these totals (roughly a fifth) matches the survey-based figure in CEP (2025).

Denominator. Division total turnover from the ONS Annual Business Survey (non-financial business economy, Section C sheet), taken at each division's latest non-suppressed year (2023 for most divisions; beverages fall back to 2022 owing to confidentiality suppression). Turnover proxies gross output at basic prices; the intensity is US exports over turnover. Divisions with no US exports or suppressed turnover are omitted and take a zero shock.

B Shock mechanics

Displacement draw. For each exposed division j, the head-count quota is $s_j \times W_j$, where W_j is the division's weighted employee count. Division members are ordered by uniform random keys—so a represented worker's inclusion probability does not depend on their record's grossing weight—and survey weights are consumed against the quota in that order. The quota-crossing record is included with probability equal to the remaining quota fraction of its weight, so the expected displaced weight equals the quota exactly. Displaced workers move to employment status UNEMPLOYED with employment income, hours, employee and salary-sacrifice pension contributions, and statutory maternity, paternity and sick pay all zeroed—without which a displaced worker would remain classified in-work (retaining, for example, childcare entitlements) or keep deducting pension contributions from zero earnings. The pipeline verifies that every displaced person's employment status actually changed in the shocked simulation and raises a hard error otherwise: a silently rejected input would alter entitlements in every downstream result.

Wage-cut earnings-equivalence. The displacement quota removes fraction s_j of division j's weighted head count with ordering keys independent of earnings, so the expected earnings removed is $s_j B_j$, with B_j the division wage bill, up to within-draw selection covariance. The wage-cut family cuts every employee in division j by exactly s_j, removing $\sum_j s_j B_j$ identically—the same expected aggregate loss, delivered on the intensive margin, with conservation asserted at run time rather than assumed. The margin comparison therefore holds the aggregate gross earnings loss fixed by construction.

Inactivity margin. The displacement draw, with displaced workers aged 50 and over (the threshold is a scenario parameter) transitioned to OTHER_INACTIVE rather than UNEMPLOYED, verified with the same hard-error contract. As discussed in Section 4.4, in policyengine-uk 2.89.2 this status distinction does not yet change computed entitlements—no UC health element or conditionality modelling—so inactivity results equal displacement results; the transition machinery is in place for the health-element extension. At seed 0 under the full schedule, 33 per cent of the displaced are 50 or over, so the margin's eventual scope is roughly a third of the displaced population.

C Monte Carlo details and manual measures

Draws and seeds. Each displacement scenario is run for 50 draws with seeds $0, \ldots, 49$; reported values are means with sample standard deviations ($n{-}1$). Per-draw decile and regional profiles are stored for every draw, and the decile, regional and age-band numbers in the text are Monte Carlo means with SDs across the 50 draws. The wage-cut margin is deterministic and collapses to a single run; the inactivity margin duplicates displacement in the current model version (Appendix B), so its results reuse the displacement draws rather than recomputing them. Draw-to-draw dispersion is substantial—the Exchequer-cost SD is about £0.30 billion against means of £0.7–0.9 billion—because with roughly 30,000 displaced workers nationally, single high-weight, high-earnings FRS records move aggregate outcomes; the text applies the discipline of narrating only orderings that survive the SD. Full and EPD displacement scenarios share seeds, so paired-draw differences partially difference out the draw noise (the full−EPD Exchequer difference is £0.19 billion $\pm$ 0.38 and positive in 36 of 50 paired draws; the deterministic wage-cut Exchequer cost exceeds the displacement draw's in 37 of 50 draws under each schedule).

Absolute poverty with a fixed line. The installed model version exposes relative BHC and AHC poverty but no absolute-poverty variable. Absolute poverty changes are therefore computed manually: the baseline BHC poverty line (60 per cent of the *baseline* equivalised median) is held fixed, and the shocked population is evaluated against it, so the measure captures absolute impoverishment rather than movement of the line itself. At seed 0 the fixed-line changes are +0.058 percentage points (full displacement/inactivity), +0.029 (EPD displacement/inactivity) and zero on the wage-cut margin under either schedule—close to the relative-line changes, confirming the shock is too small to move the national median materially.

Cushioning accounting. The cushioning rate of Section 4.2 is $1 - \Delta Y/\Delta E$, where ΔE is the weighted gross employment-earnings loss and ΔY the weighted aggregate fall in HBAI household disposable income, both at seed 0. Because it is a ratio taken within a draw, draw noise in the numerator and denominator largely cancels—which record is drawn moves gross and net losses together—so the 33-versus-41 per cent displacement/wage-cut comparison is far less draw-sensitive than the Exchequer-cost levels; the same ordering holds under both tariff scenarios (32.8 vs 41.0 full; 35.1 vs 40.8 EPD).

D Sensitivity grid: elasticity and pass-through

The derived shock (Equation 1) depends on the elasticity ε and the pass-through π only through their product $\varepsilon\pi$, so the grid in Table 8—$\varepsilon \in \{1.0, 1.5, 2.0, 3.0\} \times \pi \in \{0.5, 0.75, 1.0\}$, full schedule, displacement (10 draws per cell, seeds 0–9) and the deterministic wage cut—has a predicted structure, which the computed cells confirm. First, *levels are linear in $\varepsilon\pi$*: the wage-cut gross earnings loss per unit of $\varepsilon\pi$ is £0.886 billion in every cell to machine precision (no cell's sector shock reaches the clip at 100 per cent of the wage bill), and cells with equal products are identical—for example $(\varepsilon, \pi) = (2.0, 0.5)$ reproduces $(1.0, 1.0)$ exactly. The displacement gross loss per unit of $\varepsilon\pi$ varies only within 13 per cent across cells, which is quota rounding and draw noise, not curvature. Second, *rates are invariant*: the displacement cushioning rate stays within 32.9–33.9 per cent and the wage-cut rate within 40.6–41.5 per cent across a six-fold range of shock sizes, and the wage-cut rate exceeds the displacement rate in all twelve cells. The paper's structural claims—the cushioning gap, its decomposition, and every ordering—are therefore properties of the tax–benefit rules, not of the calibrated $\varepsilon = 2.0$; only the pound-denominated levels (and the near-proportional poverty effect) move with $\varepsilon\pi$, and they move linearly, so readers preferring a different anchor can rescale directly.

Table 8: Sensitivity grid over $\varepsilon \times \pi$, full tariff schedule

			Displacement (10 draws, mean ± SD)					Wage cut	
ε	π	$\varepsilon\pi$	Gross loss (£bn)	Exchequer (£bn)	Cushion (%)	Displaced (000s)	Poverty (pp)	Gross loss (£bn)	Cushion (%)
1.0	0.50	0.50	0.40	0.16 ± 0.10	32.9 ± 5.2	7.9	+0.014	0.44	41.0
1.0	0.75	0.75	0.69	0.29 ± 0.09	33.9 ± 4.4	13.2	+0.022	0.66	41.5
1.0	1.00	1.00	0.87	0.37 ± 0.11	33.9 ± 3.5	17.2	+0.030	0.89	41.3
1.5	0.50	0.75	0.69	0.29 ± 0.09	33.9 ± 4.4	13.2	+0.022	0.66	41.5
1.5	0.75	1.13	0.94	0.40 ± 0.11	33.9 ± 3.3	18.6	+0.031	1.00	41.3
1.5	1.00	1.50	1.32	0.55 ± 0.14	33.0 ± 2.8	26.5	+0.044	1.33	41.3
2.0	0.50	1.00	0.87	0.37 ± 0.11	33.9 ± 3.5	17.2	+0.030	0.89	41.3
2.0	0.75	1.50	1.32	0.55 ± 0.14	33.0 ± 2.8	26.5	+0.044	1.33	41.3
2.0	**1.00**	**2.00**	1.68	0.70 ± 0.16	32.9 ± 2.6	32.9	+0.056	1.77	41.0
3.0	0.50	1.50	1.32	0.55 ± 0.14	33.0 ± 2.8	26.5	+0.044	1.33	41.3
3.0	0.75	2.25	2.04	0.87 ± 0.23	33.5 ± 4.6	40.4	+0.063	1.99	40.9
3.0	1.00	3.00	2.62	1.10 ± 0.26	33.1 ± 4.3	52.1	+0.084	2.66	40.6

Notes: full tariff schedule, period 2026. Displacement cells are means ± SD over 10 Monte Carlo draws (seeds 0–9); the central calibration ($\varepsilon = 2.0$, $\pi = 1.0$, bold) differs slightly from the 50-draw headline numbers because of the smaller draw count. The wage-cut margin is deterministic; its small cushioning-rate variation (40.6–41.5 per cent) reflects the sector composition of the marginal shock, not $\varepsilon\pi$. Cells with equal $\varepsilon\pi$ are identical by construction. Poverty is the relative BHC rate change in percentage points. Displaced counts are weighted workers in thousands.

E Duration and take-up sensitivities

Table 9 reports two sensitivities on the full-schedule displacement margin, each over the same 10 draws (seeds 0–9) so columns are directly comparable to the recomputed central case.

Out-of-work duration. The central results annualise a twelve-month spell. Under a six-month expected duration, displaced workers keep 50 per cent of baseline annual earnings, evaluated *in-model* so taxes and means-tested benefits respond to the actual annual income (a documented hybrid, following the sister study's convention: the annual model has no intra-year timing, so the displacement transition—hours zeroed, status unemployed—is unchanged while half a year's earnings are retained). The gross loss halves by construction; the annualised Exchequer cost falls roughly pro rata, in fact slightly more than half (£0.70 to £0.41 billion, 59 per cent of the full-year figure) because the retained earnings slice is taxed above the displaced worker's average rate; the cushioning rate rises from 33 to 42 per cent for the same marginal-versus-average-rate reason that drives the paper's headline gap; and the poverty effect nearly vanishes—half a year of earnings keeps most displaced households above the line.

UC take-up. PolicyEngine models UC take-up stochastically through the benunit-level `would_claim_uc` dataset input (the mechanism decomposition of Section 4.3 found 55 per cent of zero-UC displaced workers are in families modelled as not taking up). Forcing 100 per cent take-up in *both* the baseline and the shocked simulation—so the deltas are shock effects under full take-up, not take-up effects—more than doubles the UC response (£0.16 versus £0.07 billion) and raises the displacement cushioning rate to 38.6 ± 2.5 per cent, closing about seventy per cent of the displacement–wage-cut gap. Modelled non-take-up is thus the single largest reason UC fails to reach the displaced; the remainder is chiefly the joint partner-income taper.

Table 9: Duration and take-up sensitivities, full-tariff displacement (10 draws)

	Central (12-month, modelled take-up)	Duration 6 months	Take-up 100 per cent
Gross earnings loss (£bn/yr)	1.68 ± 0.32	0.84 ± 0.16	1.68 ± 0.32
Exchequer cost (£bn/yr)	0.70 ± 0.16	0.41 ± 0.09	0.79 ± 0.16
Cushioning rate (%)	32.9 ± 2.6	42.2 ± 3.3	38.6 ± 2.5
UC response (£bn/yr)	0.07 ± 0.03	0.01 ± 0.01	0.16 ± 0.06
Poverty change, BHC (pp)	$+0.056 \pm 0.014$	$+0.013 \pm 0.010$	$+0.048 \pm 0.019$

Notes: means ± SD over 10 Monte Carlo draws (seeds 0–9), shared across columns. The take-up column compares a forced-take-up shocked simulation against a forced-take-up baseline (person-weighted modelled take-up in the dataset: 55 per cent). The duration column's cushioning rate is computed against its own halved gross loss.

F Regional incidence table

Table 10 reports the full three-scenario regional profile; Figure 5 maps its two displacement columns at ITL1/government office region level (the constituency-level counterpart is Figure 4 in Section 4.8).

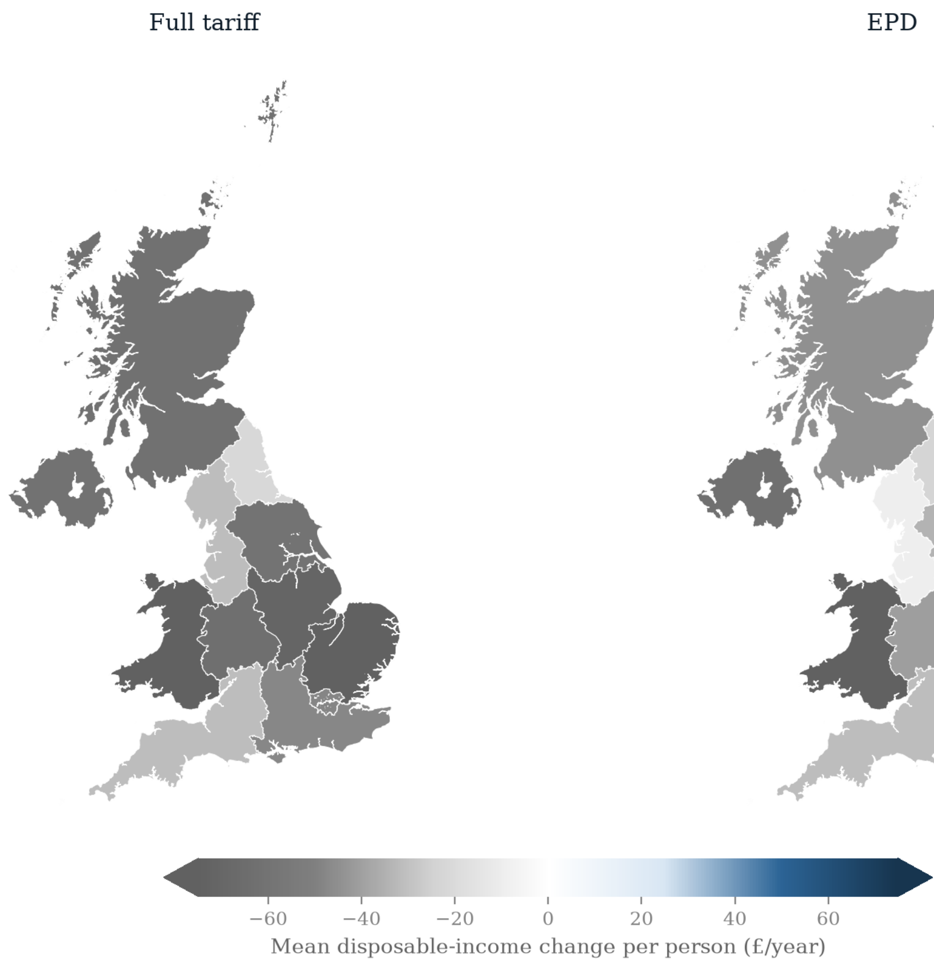

Figure 5: Regional choropleth maps of the mean disposable-income change per person per year (£, Monte Carlo mean over 50 displacement draws), by ITL1/government office region: full-tariff displacement (left) and EPD displacement (right), on a shared diverging scale (losses in grey). The diverging scale is clipped at the 95th percentile of the absolute change (caps show clipping). Region-level draw-to-draw SDs are comparable to the means (Table 10), so the map shows the average geography, not a sharply identified ranking.

Table 10: Mean fall in household disposable income per person per year by region (£, Monte Carlo means ± SD over 50 draws)

	Full tariffs		EPD
Region	Displacement	Wage cut	Displacement
Wales	108 ± 234	65	75 ± 181
East of England	74 ± 76	61	51 ± 59
East Midlands	72 ± 73	51	70 ± 93
West Midlands	64 ± 56	55	40 ± 54
Scotland	62 ± 50	39	45 ± 37
Northern Ireland	61 ± 46	49	63 ± 52
Yorkshire and the Humber	60 ± 48	47	34 ± 37
London	48 ± 53	52	46 ± 49
South East	47 ± 40	42	41 ± 37
North West	32 ± 34	27	10 ± 13
South West	31 ± 31	36	31 ± 28
North East	22 ± 47	18	23 ± 46

Notes: absolute values of the per-person disposable income change; displacement columns are Monte Carlo means ± draw-to-draw SD over 50 draws, the wage-cut column is the deterministic run. SDs are comparable to or larger than the means—no pairwise regional ordering survives one SD; the Wales mean is dominated by a single high-weight FRS record (Appendix C).